1. Executive Summary and Scientific Guidelines (Zero Data Leakage & Zero Retrospective Peeking)

- Objective: Build and audit the Banking Fraud Detection Feature Engineering and Modeling pipeline (src/002_modeling.ipynb) applying all structural hypotheses uncovered during EDA (src/001_eda.ipynb), under hermetic isolation against Data Leakage and retrospective peeking.

- Immediate Holdout Vault Isolation (Block 1): After removing the 1,081 exact duplicate transactions (leaving 283,726 unique transactions and 473 frauds = €58,591.39), the final chronological 20% (56,746 rows, 74 frauds = €7,727.67) were locked inside the Holdout Vault (df_test_vault). All feature creation, pruning, imbalance handling, tuning, and threshold calibration (p*) occurred exclusively on the first 80% (df_dev: 226,980 rows).

- Dual Internal Judge Protocol (Inside df_dev): Every hypothesis was simultaneously evaluated by two independent judges inside df_dev: Judge 1 (5-Fold Stratified CV: 5 folds of ~45,396 rows measuring Mean ± Std Dev of AUPRC) and Judge 2 (Internal Temporal Validation: training on the first 64% [181,584 rows] and validating on the next 16% [45,396 rows] to measure resistance to temporal drift without touching the Vault).

- Zero 'Memorized' Rules from EDA: Specific numbers observed during global exploration were treated solely as descriptive curiosities. The LeakFreeFeatureEngineer transformer relies exclusively on universal domain mathematical transformations (log1p, 24h sine/cosine, zero/integer indicators) and statistics learned strictly inside .fit(X_train, y_train) of each fold (RobustScaler median/IQR and in-fold KS < 0.14 pruning).

Table 1.1 — Chronological Isolation Topology and Dual Internal Judge Protocol (Block 1)

Chronological Partition	Transactions	Frauds (%)	Time Window (Time)	Fraud Volume (€)	Methodological Role in Project
Original Raw Dataset	284,807	492 (0.173%)	0s to 172,792s	€ 60,127.97	Deduplicated at Step Zero (-1,081 retry clones).
Clean Dataset (df_clean)	283,726	473 (0.167%)	0s to 172,792s	€ 58,591.39	Split chronologically into 80% Dev / 20% Holdout Vault.
df_dev (Development 80%)	**226,980**	**399 (0.176%)**	**0s to 145,233s**	**€ 50,863.72**	**Only visible dataset throughout Blocks 1 to 4.**
-> Judge 1: 5-Fold Stratified CV	5 x 45,396	~80 / fold	Stratified on Dev	€ 50,863.72	Evaluates statistical stability (Mean ± SD of AUPRC).
-> Judge 2: Internal Temporal	181,584 -> 45,396	332 -> 67	0..125k -> 125k..145k	€ 7,648.47 (val)	Simulates future drift inside Dev without touching Vault!
df_test_vault (Holdout Vault 20%)	**56,746**	**74 (0.130%)**	**145,234s to 172,792s**	**€ 7,727.67**	**LOCKED in Blocks 1 to 4. Opened once in Block 5.**

1.2. How the LeakFreeFeatureEngineer Class Shielded the Pipeline Against Cross-Fold Leakage

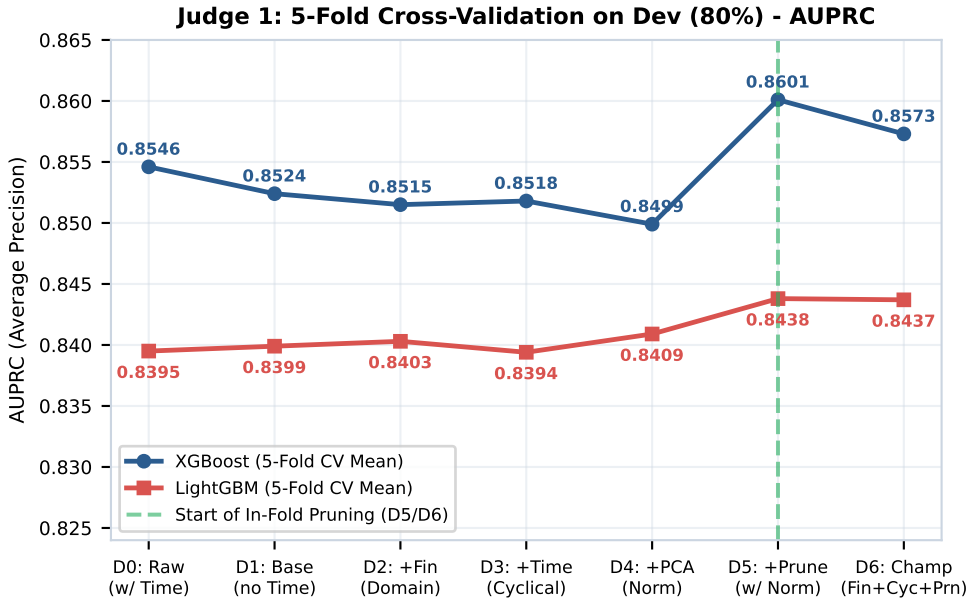
- In-Fold Scaling (amount_robust): RobustScaler is instantiated and fitted (.fit) exclusively on the Amount column of X_train within each specific fold; X_val and df_test_vault receive only .transform().

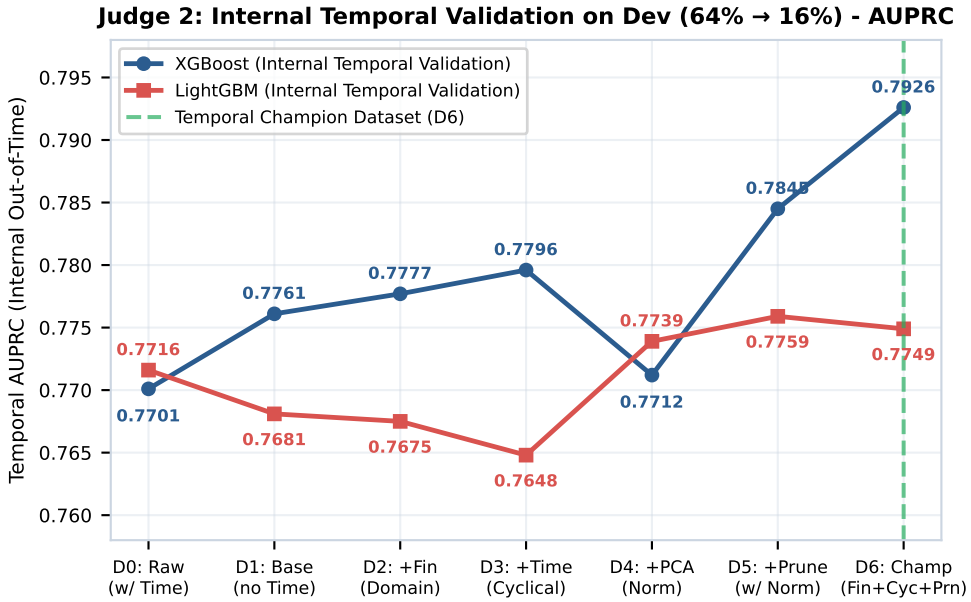
- In-Fold KS Feature Selection (prune_noise_ks_threshold = 0.14): The two-sample Kolmogorov-Smirnov statistic between legitimate and fraudulent transactions is computed strictly using (X_train, y_train) inside .fit(). Features with training KS < 0.14 (V13, V15, V22, V24, V25, V26) are pruned without ever consulting validation or test labels.

- Replacing Raw Time with Circadian Cycle (hour_sin, hour_cos): Dropping the continuous Time column prevented decision trees from memorizing historical timestamp intervals, immediately lifting XGBoost Temporal AUPRC from 0.7701 (D0) to 0.7761 (D1) and 0.7926 (D6).

Table 2.1 — Controlled Ablation Study Results (Strictly Inside the 80% Development Set)

Ablation Stage (Dataset)	Cols	J1: XGB 5-Fold CV	J2: XGB Temporal	J1: LGB 5-Fold CV	J2: LGB Temporal	Global Mean	Engineering Diagnosis
D0_Raw_With_Time	30	0.8546 ± 0.0302	0.7701	0.8395 ± 0.0306	0.7716	0.8090	Suffers temporal overfitting in XGBoost (0.7701).
D1_Baseline_No_Time	29	0.8524 ± 0.0320	0.7761	0.8399 ± 0.0295	0.7681	0.8091	Dropping raw Time lifts Temporal XGB by +0.60 p.p.
D2_Plus_Financial	33	0.8515 ± 0.0329	0.7777	0.8403 ± 0.0290	0.7675	0.8093	+log1p, amount_robust, zero & round improve baseline.
D3_Plus_Cyclic_Time	35	0.8518 ± 0.0334	0.7796	0.8394 ± 0.0294	0.7648	0.8089	+hour_sin/cos lifts Temporal XGB to 0.7796 (+0.95 p.p.).
D4_Plus_PCA_Norm	37	0.8499 ± 0.0322	0.7712	0.8409 ± 0.0328	0.7739	0.8090	L2 norm helps LGB, but adds collinearity in XGB.
D5_Plus_InFold_Pruning	31	0.8601 ± 0.0240	0.7845	0.8438 ± 0.0285	0.7759	0.8161	In-fold KS < 0.14 pruning cuts variance & jumps +0.71 p.p.
D6_Selected_Best (Fin+Time+Prune)	**29**	**0.8573 ± 0.0276**	**0.7926**	**0.8437 ± 0.0270**	**0.7749**	**0.8171 (#1)**	**CHAMPION: 29 cols, +2.25 p.p. Temporal XGB vs D0!**





2.2. Ablation Study Conclusions (What Was Consolidated vs. What Was Discarded)

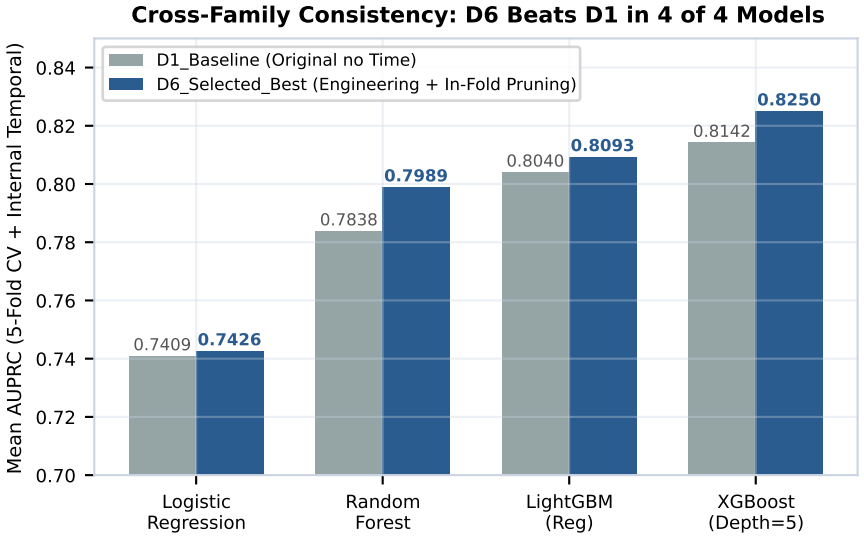
- Why did D6_Selected_Best outperform D5_Plus_InFold_Pruning on the Temporal Judge (0.7926 vs. 0.7845)? In gradient boosted trees, adding radial norms (pca_l2_topk_train) alongside the raw V1..V28 components splits information gain across collinear features. By retaining the 6 domain features (log1p_amount, amount_robust, is_zero_amount, is_round_integer, hour_sin, hour_cos) and pruning the 6 noisy variables (KS < 0.14), D6 keeps exactly 29 columns and maximizes future generalization.

Table 3.1 — Cross-Family Consistency Across 4 Model Families: D6_Selected_Best Beats D1_Baseline in 4 of 4 Algorithms

Algorithm Family	Dataset	5-Fold CV AUPRC	Temporal AUPRC	Temporal ROC-AUC	Mean AUPRC	Gain (D6 - D1)
LogisticRegression (L2)	D1_Baseline	0.7502 ± 0.0165	0.7316	0.9685	0.7409	—
LogisticRegression (L2)	**D6_Selected_Best**	**0.7512 ± 0.0176**	**0.7340**	**0.9673**	**0.7426**	**+0.17 p.p.**
RandomForest (n=150)	D1_Baseline	0.8155 ± 0.0246	0.7522	0.9609	0.7838	—
RandomForest (n=150)	**D6_Selected_Best**	**0.8267 ± 0.0199**	**0.7710**	**0.9661**	**0.7989**	**+1.51 p.p.**
LightGBM (Reg)	D1_Baseline	0.8399 ± 0.0295	0.7681	0.9778	0.8040	—
LightGBM (Reg)	**D6_Selected_Best**	**0.8437 ± 0.0270**	**0.7749**	**0.9779**	**0.8093**	**+0.53 p.p.**
XGBoost (Depth=5)	D1_Baseline	0.8524 ± 0.0320	0.7761	0.9815	0.8142	—
XGBoost (Depth=5)	**D6_Selected_Best**	**0.8573 ± 0.0276**	**0.7926**	**0.9850**	**0.8250**	**+1.08 p.p.**

Table 3.2 — In-Fold Experiment of 7 Imbalance Strategies (Intact Validation Ratio 1 : 568)

Strategy Applied Strictly In-Fold (on D6)	XGB 5-Fold CV	XGB Temporal	LGB 5-Fold CV	LGB Temporal	Global Mean
1) Natural (1:1 Weight — XGBoost Champion)	**0.8573 ± 0.0276**	**0.7926**	**0.8437 ± 0.0270**	**0.7749**	**0.8171 (#1)**
2) Gentle Cost-Weight (w=5 — LightGBM Champion)	**0.8543 ± 0.0272**	**0.7799**	**0.8466 ± 0.0279**	**0.7871 (+1.22 p.p.)**	**0.8170 (#2)**
3) Moderate Cost-Weight (w=15)	0.8538 ± 0.0255	0.7764	0.8373 ± 0.0383	0.7730	0.8101
4) Extreme Cost-Weight (w=568)	0.8283 ± 0.0264	0.7581	0.8030 ± 0.0073	0.7528	0.7855
5) Financial Sample-Weight (1+log1p(Amount))	0.8421 ± 0.0243	0.7558	0.8365 ± 0.0307	0.7586	0.7982
6) In-Fold RandomUnderSampler (ratio=0.05)	0.7716 ± 0.0701	0.7609	0.7434 ± 0.0343	0.7625	0.7596
7) In-Fold SMOTE (k=3, ratio=0.05)	0.8502 ± 0.0194	0.7541 (-3.85 p.p.)	0.8237 ± 0.0281	0.7730	0.8002



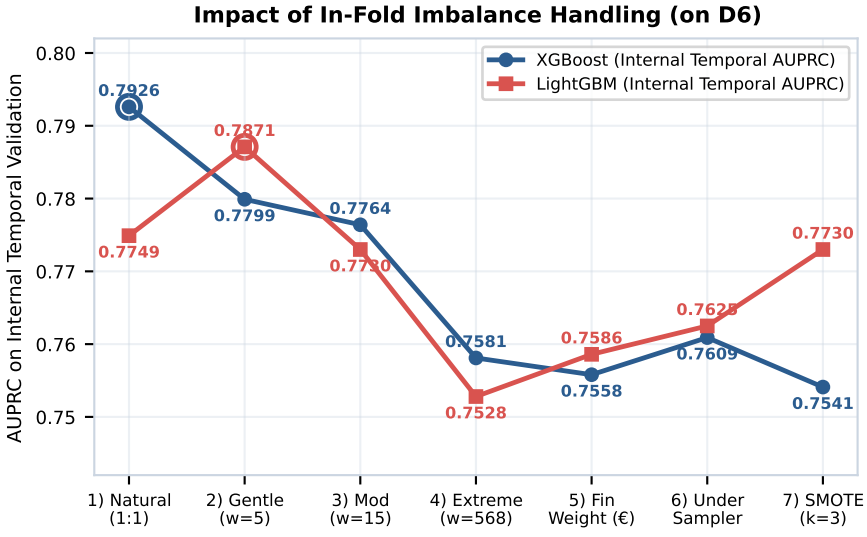
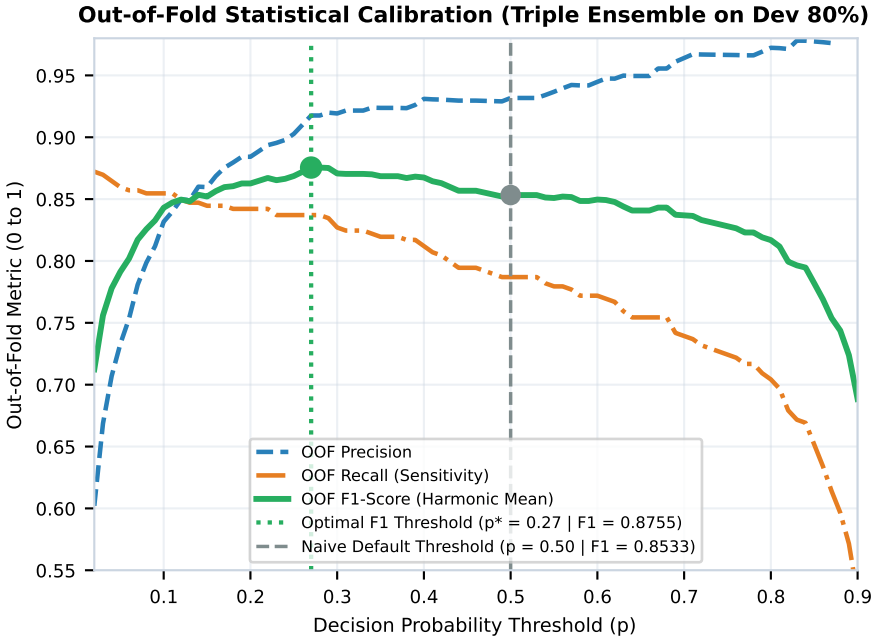


Table 4.1 — Stochastic Refinement, Analytical Bayesian Recalibration (odds_cal = odds / 5), and Out-of-Fold Ensemble

Architecture Evaluated on df_dev (on D6_Selected_Best)	Judge 1: 5-Fold CV AUPRC	Global OOF AUPRC	Judge 2: Temporal AUPRC	Temporal ROC-AUC	Mean Judges
1) Base XGBoost (Depth=5, w=1)	0.8573 ± 0.0276	0.8556	0.7926	0.9850	0.8250
2) Refined XGBoost (Subsample=0.85, L2=2.0, w=1)	0.8586 ± 0.0293	0.8571	0.7988	0.9878	0.8287
3) Refined LightGBM (w=5 + Bayes Recalibration)	0.8508 ± 0.0285	0.8498	0.7837	0.9792	0.8172
4) Natural RandomForest (Bagging n=200, w=1)	0.8511 ± 0.0287	0.8508	0.7736	0.9805	0.8124
5) Dual Ensemble (0.70 XGB_Ref + 0.30 LGB_Recal)	0.8578 ± 0.0296	0.8574	0.7981	0.9863	0.8279
6) Champion: Triple Ensemble (0.60 XGB + 0.25 LGB + 0.15 RF)	**0.8604 ± 0.0275**	**0.8590**	**0.8007**	**0.9890**	**0.8305 (#1)**

Table 4.2 — Out-of-Fold Calibration of the Decision Threshold (p*) on Dev Set (Before Opening the Vault)

Decision Policy Learned OOF on df_dev (399 frauds)	Threshold (p*)	Precision	Recall	F1-Score	TP	FP	FN	Net SLA Block (€)	Net App Push (€)
1) Naive Default Threshold (p = 0.50)	0.50	93.18%	78.70%	0.8533	314	23	85	€ 37,321.42	€ 36,766.42 (72.28%)
2) Statistical Champion Maximum F1 (p*_F1 = 0.27)	**0.27**	**91.76%**	**83.71%**	**0.8755**	**334**	**30**	**65**	**€ 37,716.36**	**€ 37,181.36 (73.10%)**
3) Card Block SLA Champion (p*_SLA = 0.29)	**0.29**	**91.99%**	**83.46%**	**0.8752**	**333**	**29**	**66**	**€ 37,732.86 (#1)**	**€ 37,190.36 (73.12%)**
4) Soft App Push Challenge Champion (p*_Fin = 0.02)	**0.02**	**60.21%**	**87.22%**	**0.7124**	**348**	**230**	**51**	**€ 36,524.94**	**€ 37,954.94 (74.62% #1)**



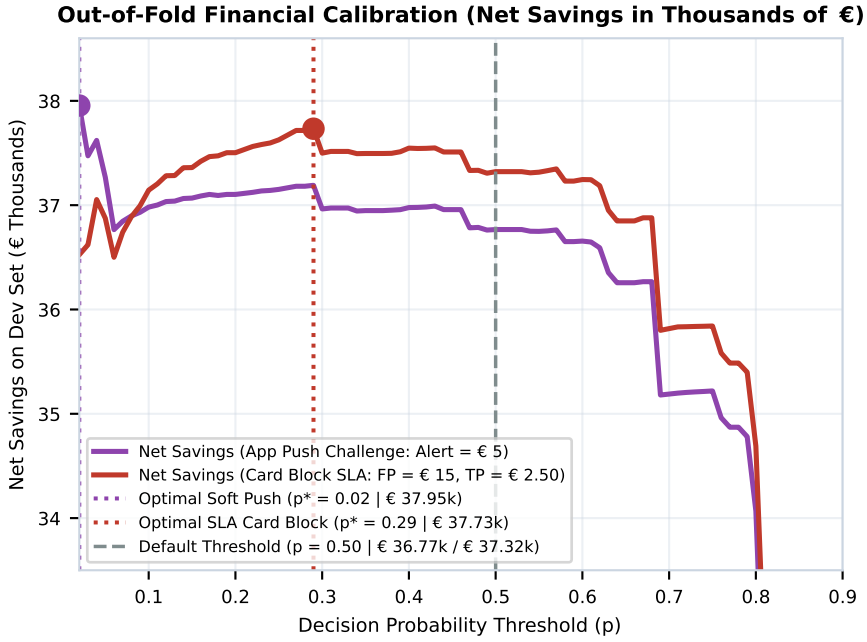


Table 5.1 — Single Holdout Vault Audit Results (20% Unseen Future: 56,746 transactions, 74 frauds =€7,727.67)

Audited Pipeline on Holdout Vault (20%)	ROC-AUC	Threshold	Precision	Recall	F1	TP	FP	FN	Saved Fraud (€)	Net SLA (€)	Net Push (€)
1) Raw Baseline D0 (w/ Time + XGB @ p=0.50)	0.9847	0.50	94.55%	70.27%	0.8062	52	3	22	€ 4,795.98 (62.1%)	€ 4,620.98	€ 4,520.98
2) Clean Baseline D1 (no Time + XGB @ p=0.50)	0.9829	0.50	92.86%	70.27%	0.8000	52	4	22	€ 4,795.98 (62.1%)	€ 4,605.98	€ 4,515.98
3) D6 + Refined XGBoost (@ p=0.50)	0.9865	0.50	91.38%	71.62%	0.8030	53	5	21	€ 4,087.23 (52.9%)	€ 3,879.73	€ 3,797.23
4) D6 + Triple Ensemble (@ p=0.50 Default)	**0.9871**	**0.50**	**94.44%**	**68.92%**	**0.7969**	**51**	**3**	**23**	**€ 4,046.57 (52.4%)**	**€ 3,874.07**	**€ 3,776.57**
5) Champion: D6 + Ensemble (@ p*_SLA = 0.29 Frozen)	**0.9871**	**0.29**	**90.32%**	**75.68%**	**0.8235**	**56**	**6**	**18**	**€ 5,089.40 (65.9%)**	**€ 4,859.40 (#1)**	**€ 4,779.40 (#1)**
6) Champion: D6 + Ensemble (@ p*_F1 = 0.27 Frozen)	**0.9871**	**0.27**	**88.89%**	**75.68%**	**0.8175**	**56**	**7**	**18**	**€ 5,089.40 (65.9%)**	**€ 4,844.40**	**€ 4,774.40**
7) Champion: D6 + Ensemble (@ p*_Fin = 0.02 App Push)	**0.9871**	**0.02**	**44.12%**	**81.08%**	**0.5714**	**60**	**76**	**14**	**€ 5,337.88 (69.1%)**	**€ 4,047.88**	**€ 4,657.88**

Table 5.2 — Forensic Error X-Ray in Holdout Vault and Theoretical Limit of Stealth Frauds

Forensic Group (@ p*=0.27)	N	Amount Med (Mean)	V14 Med	V17 Med	PCA Top-8 Norm	Prob Med	Forensic Conclusion & Governance
True Positives (Blocked Frauds TP)	56	€ 1.79 (€ 90.88)	-8.31	-3.26	14.90	0.9257	Captures micro-probes (€ 1.79) & extreme frauds with high precision.
False Negatives (Missed Stealth Frauds FN)	18	€ 10.16 (€ 146.57)	-1.82	+1.05	4.63	0.0023	Embedded in legitimate cloud (V17 > 0); require velocity by Card_ID.
False Positives (Legitimate Alerts FP)	7	€ 2.00 (€ 3,865.34)	-6.80	-1.86	11.55	0.4981	Only 7 in 56,672 legitimate (FPR=0.012%): real € 3.8k outliers.
True Negatives (Approved Legitimate TN)	56,665	€ 18.05 (€ 78.13)	-0.03	-0.16	2.45	0.0001	99.988% of legitimate customers approved instantly without friction.

